

EB PROPRIETARY

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Title: ML for Trade Inspection Report (TIR) Liability Coding

CATT

Sponsor(s):

CURRENT STATE

There has been an increase in UNSATs between key nuclear events. There is also an increase in the time associated with adjudication and resolution of UNSATs. Electric Boat, D424, currently has a team of three full time resources which process upwards of 15 data types (surveillance UNSATs, CARs, events, etc.), with TIRs being the largest by volume, to create quality and insight tracking reports which in turn help operations as well as others throughout EB. The team currently cleans, characterizes, and codes data received while typically processing an average of 1000 records, of various types per week, with 40,000+ records coded and validated yearly. Coding of TIR liabilities to support long term quality trending within the shipyard is labor intensive, tedious and inherently introduces a level of subjectivity based on experience.

AREA OF RESEARCH

- QPS Data
- TIR Liability Coding Process
- Machine Learning Statistical Analysis
- Small Language Models
- Large Language Models

PROJECT GOALS

1. Investigate QPS TIR records to identify if data is sufficiently labeled to support the training of a machine learning model.
2. Determine if QPS data can easily be exported.
3. Explore the TIR coding process to understand how coding occurs with trade inspection inputs (i.e. 3-5% of TIR inputs require additional investigation).
4. Assess the existing ML for TIR Liability Coding project plan provided by GDMS for viability based on data findings within QPS.
5. Establish a benchmark for success for a machine learning solution (i.e. TIRs coded with X% accuracy or Y% time savings when coding TIRs).

POTENTIAL BENEFITS

- Machine learning may provide a means to assist coders with TIRs which may result in potential span reduction.
 - Coded unsat data would be available sooner to the stakeholders.
 - Reduce workload of dedicated liability coders allowing for renewed focus on root cause analysis and long term quality trending.

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